

Se tiene un vector $(v_0, v_1, \dots, v_{(n-1)})$ deseo calcular el promedio móvil, que consiste en tomar k elementos y calcular su promedio, por ejemplo

$$(1, 2, 3, 4, 5) \quad k = 3$$

$$(1+2+3)/3 = 2$$

$$(2+3+4)/3 = 9/3 = 3$$

$$(3+4+5)/3 = 12/3 = 4$$

$$T = 2$$

$$(1+2+3)/3, \quad (2+3+4)/3$$

$$T2 \quad (3+4+5)/3$$

$$T = 3$$

$$(1+2+3)/3$$

$$(2+3+4)/3$$

$$(3+4+5)/3$$

Si mi vector es tamaño n , y tengo una ventana tamaño k
 $(v_0, v_1, \dots, v_{n-1})$

$$\left\{ \begin{array}{ll} v_{k-1} & 0, 1, 2, 3, \dots, v_{k-1} \\ v_k & 1, 2, 3, 4, \dots, v_k \\ v_{k+1} & 2, 3, 4, \dots, v_{k+1} \\ \dots & \\ v_{n-1} & v_{n-1-k}, v_{n-1-(k+1)}, \dots, v_{n-1} \end{array} \right.$$

$$(v_0, v_1, v_2, v_3, v_4) \quad \underline{k = 3}$$

$$0, 1, 2 \quad (v_{k-1}) \quad v_0$$

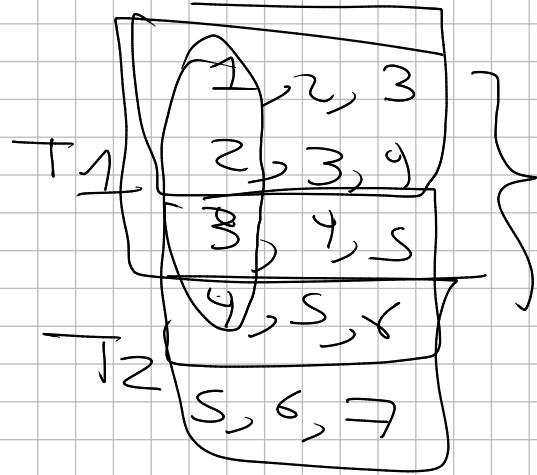
$$1, 2, 3$$

$$2, 3, 4 \quad (v_{n-1})$$

1, 2, 3, 4, 5, 6, 7

$$n = 7$$

$$k = 3$$



1, 2, 3, 4, 5, 6, ... n

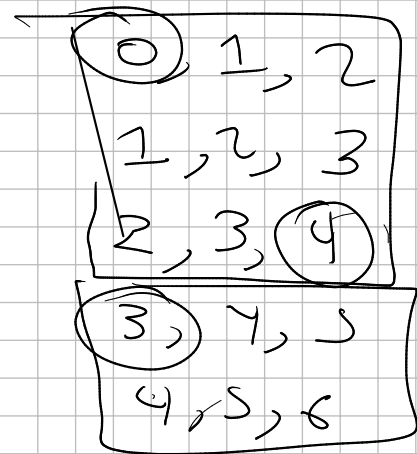
$$n = \left\lceil \frac{7}{2} \right\rceil = 4$$

$$n - \left\lceil \frac{7}{2} \right\rceil = 3$$

$$i = 0, \quad \sqrt{n}$$

0, 1, 2, 3, 4, 5, 6, ...

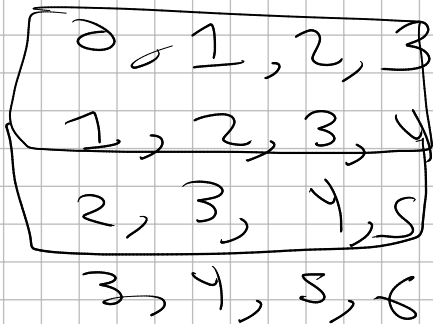
$$k = 3$$



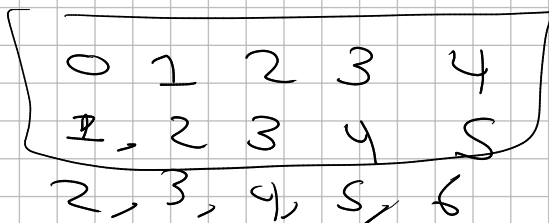
$$i = 0, \quad k + 1$$

$$k, \quad n - 1$$

$$k = 4$$



$$k = 5$$



$$0, \quad k$$

$$n = 7$$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

$$K=3$$

$$K=3$$

$$n=10$$

0	1, 2
1	2, 3
2	3, 4
3	4, 5

4	5, 6
5	6, 7
6	7, 8
7	8, 9

[0 ... 5]

[4 ... 9]

$$n=10$$

$$K=3$$

I = 0, 1, 2, 3, 4, 5, 6, 7

0 ————— n - K

$$0 - \frac{n-K}{2}, \frac{n-K}{2} + 1, n - K$$

$$\left[0, \frac{n-K}{2} + (K-1), \frac{n-K}{2} + 1, n-1 \right]$$

$$\left[0, \frac{10-3}{2} + (3-1), \frac{10-3}{2} + 1, 10-1 \right]$$

$$[0, 3+2] [3+1, 9]$$

$$[0, 5] [4, 9]$$

necesitamos calcular el promedio movil, este consiste en tomar un vector de tamaño n , y una ventana de tamaño k

Por ejemplo $n = 5$, $k = 3$

$T = 2$

$(5, 8, 2, 6, 1)$

$$(5+8+2)/3$$

$$(8+2+6)/3$$

$$(2+6+1)/3$$

$\textcircled{0} \textcircled{1} \textcircled{2} [0 \dots 2] [1 \dots 4]$

$1, 2, 3$
 $2, 3, 4$

$n = 5 \quad k = 3$

$(0, 1, 2, 3, 4, 5, \dots, n-1)$

k

$0, 1, 2, 3, \dots, k-1$
 $1, 2, 3, 4, \dots, k$
 $\vdots$
 $n-k, \dots, n-1$

$0, 1, 2, 3, \dots, n-k$

$$\frac{n-k}{2} \mid \frac{n-k}{2} + 1$$

$$0 \dots \frac{n-k}{2} + (k-1) \mid \frac{n-k}{2} + 1, n-1$$

$n = 10 \quad k = 5 \quad T = 2$

$$[0, \frac{10-5}{2} + 4]$$

$$[0, 6] [3, 9]$$

Split At

$0, 1, 2, 3, 4$

$1, 2, 3, 4, 5$

$2, 3, 4, 5, 6$

$3, 4, 5, 6, 7$

$4, 5, 6, 7, 8$

$5, 6, 7, 8, 9$